CMS · NANOAOD · TRIGGER OBJECTS

# The unseeded diphoton leg never gets its NanoAOD filter bits

Photon **TrigObj_filterBits** 16 and 17 (14 and 15 before NanoAOD v15) are documented for HLT_Diphoton30_22_R9Id_OR_IsoCaloId_AND_HE_R9Id_Mass90 / Mass95, but no photon trigger object ever carries them. Why it happens, how widespread it is, and the one-line fix.

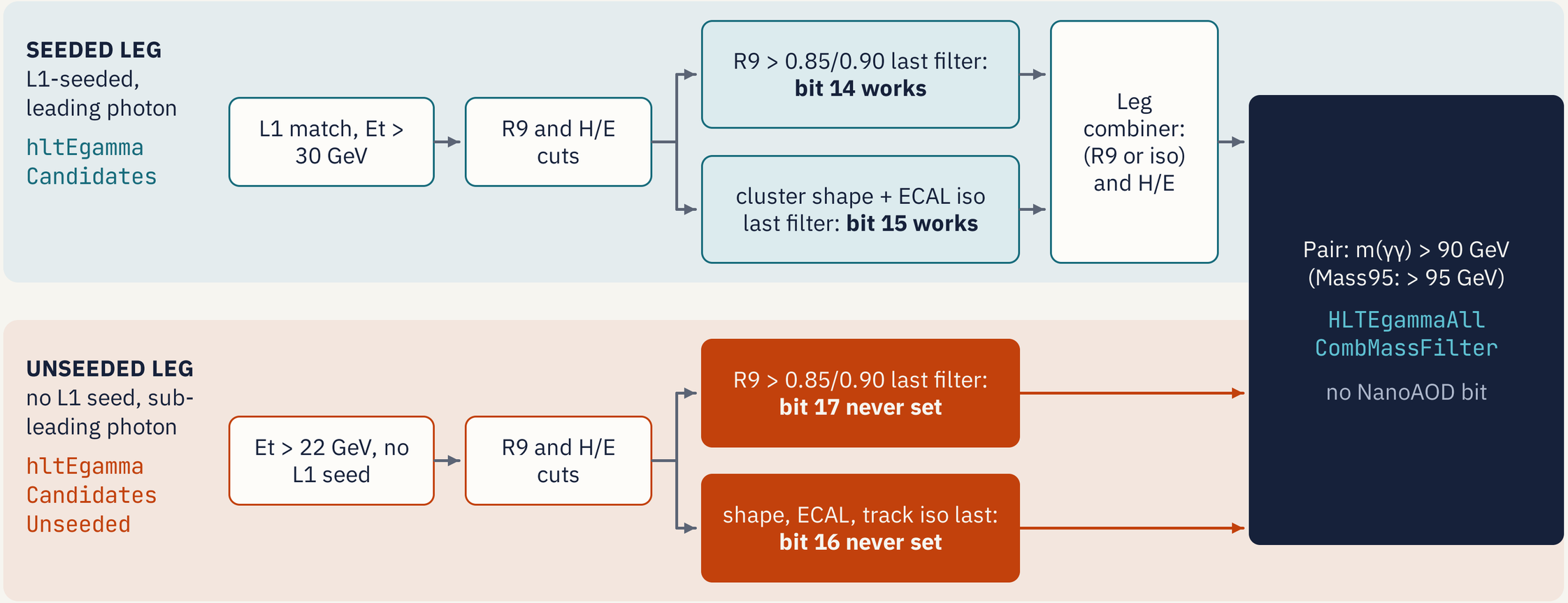
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Checked against cms-sw/cmssw master ed35654b1 (today), release branches 13_0_X to 16_1_X, and the 2018 to 2026 HLT menus

`PhysicsTools/NanoAOD/python/triggerObjects_cff.py`

# One path, two legs, four last filters

HLT_Diphoton30_22_R9Id_OR_IsoCaloId_AND_HE_R9Id_Mass90 and Mass95 (same sequence), every menu 2018 to 2026




Orange boxes: the unseeded last filters NanoAOD discards. Bits in v15 numbering; v12 to v14: 12, 13, 15, 14.

# From the HLT filter to TrigObj_filterBits: the object survives four hand-offs and dies at the fifth

| | |
|---|---|
| `1 · HLT filters (saveTags)` | The unseeded last filters tag their objects with the collection **hltEgammaCandidatesUnseeded**; the leg's first filter saves them with type 92, the later ones with type 81. |
| `2 · hltTriggerSummaryAOD` | Objects are stored once per HLT collection; every filter that accepted an object records its key and the type it saved it with. |
| `3 · patTrigger → slimmedPatTrigger (MiniAOD)` | The pat object keeps the collection label **hltEgammaCandidatesUnseeded::HLT**, all its types (92 and 81) and the labels of every filter it passed. |
| `4 · unpackedPatTrigger (NanoAOD)` | Filter labels are unpacked; the object reaches the table producer intact, with its collection name and types. |
| `5 · triggerObjectTable, Photon selection` | **sel = type(92) && pt > 15 && coll('hltEgammaCandidates')** is evaluated before the quality bits. The collection test fails, the object is dropped, and bits 16 and 17 are never computed. |

# Where it breaks: the producer's decision for one unseeded photon object

**1** **type(92)?** yes. The leg's first filter (HLT1Photon, triggerType 92) saved the object as TriggerCluster; PAT keeps every type.

**2** **pt > 15?** yes. The unseeded leg requires Et > 22 GeV (18 GeV for the Diphoton30_18 paths).

**3** **coll('hltEgammaCandidates')?** no. The stored label is hltEgammaCandidatesUnseeded::HLT. Without a wildcard the comparison is exact (TriggerObjectStandAlone::hasCollection falls back to TriggerObject::hasCollection).

**4** **Discarded before the bits.** qualityBits is only evaluated for objects that pass sel; the duplicate merging only sees accepted objects. Bits 16 and 17 stay 0 in every event.

```
# triggerObjects_cff.py, Photon
sel = "type(92) && pt > 15 &&
       coll('hltEgammaCandidates')"
skipObjectsNotPassingQualityBits = True
```

```
// TriggerObjectTableProducer.cc, produce()
for (const auto &obj : trigObjs)
  for (const auto &sel : sels_)
    if (sel.match(obj)) { // fails here
      bits = sel.qualityBits(obj);
      if (bits =/= 0) selected.push_back(...);
    }
```

Same logic in 13_0_X, 13_3_X, 14_0_X, 15_0_X and master; only the bit width (32 bits up to 15_0_1, 64 bits from 15_0_2) and the merging rule changed.

# The photon bit list: what is documented versus what can be set

| bit v15 | bit v12 to v14 | filter named by the bit | objects come from | status |
|---|---|---|---|---|
| 8 | 8 | `hltHtEcal800` | hltHtMhtEcal (HT object, type 89) | **never set**: not an e/gamma object, since v10 |
| 14 | 12 | `hltEG30LR9Id85b90eHE12R9Id50b80eR9IdLastFilter` | hltEgammaCandidates | **works** (seeded leg) |
| 15 | 13 | `hltEG30LIso60CaloId15b35eHE12R9Id50b80e EcalIsoLastFilter` | hltEgammaCandidates | **works** (seeded leg) |
| 16 | 14 | `hltEG22Iso60CaloId15b35eHE12R9Id50b80e TrackIsoUnseededLastFilter` | hltEgammaCandidatesUnseeded | **never set**: collection rejected by sel |
| 17 | 15 | `hltEG22R9Id85b90eHE12R9Id50b80eR9UnseededLastFilter` | hltEgammaCandidatesUnseeded | **never set**: collection rejected by sel |
| 18 | 16 | `hltEG30Iso60CaloId15b35eR9Id50b90e HE12b10eR9Id50b80eEcalIsoFilter` | hltEgammaCandidates | **works for 2018 and 2022 menus**; filter renamed in 2023, stale since |
| 19 | 17 | `hltEG18TrackIso60Iso60CaloId15b35eR9Id50b90e HE12b10eR9Id50b80eTrackIsoUnseededFilter` | hltEgammaCandidatesUnseeded | **never set** (collection), and renamed in 2023 |
| 20 | none | `hltEG*L1VBFLooseIsoEGHEFilter` | hltEgammaCandidates | **works** (filter exists from 2023) |

Bits 0 to 7 (single-photon HE filters), 9 to 13 (EB tight-ID filters and the Mu17_Photon30 photon leg) are fine. Decimal values of the dead unseeded bits: 65536, 131072, 524288 in v15 and master; 16384, 32768, 131072 in v12 to v14.

# How widespread (1): every NanoAOD version since v12, every menu since 2018, still there today

| NanoAOD | release checked | photon selection | dead unseeded bits | affected |
|---|---|---|---|---|
| v9 | CMSSW_10_6_X | seeded only, no photon bits at all | none defined | no photon TrigObj |
| v10, v11 | CMSSW_12_4_X, 12_6_X | seeded only, pt > 20 | bits absent | no |
| v12 | CMSSW_13_0_0 to 13_0_17 (prod 13_0_13) | seeded only | 14, 15, 17 | yes |
| v13 | CMSSW_13_3_1_patch1 | seeded only | 14, 15, 17 | yes |
| v14 | CMSSW_14_0_X, 14_1_X, 14_2_X | seeded only | 14, 15, 17 | yes |
| v15 | CMSSW_15_0_2 to 15_0_15_patch4, 15_1_X | seeded only | 16, 17, 19 | yes |
| next | 16_0_X, 16_1_X (tags to 16_1_3), master ed35654b1 today | seeded only, no open PR or issue | 16, 17, 19 | yes |

### Menus

19 menu files checked, 2018 GRun to the frozen 2026 menu: the four Diphoton30_22 filters exist everywhere, always on the same collections, with the same 16-filter chain. Path versions v13 (2018, early 2022) to v29 (2026).

### Datasets

Every Run 3 NanoAOD file: 2022 to 2025 data (PromptNano, 22Sep2023, 16Dec2023, v15 re-NANO) and the Summer22, 22EE, 23, 23BPix and 24 MC campaigns, plus any Run 2 input re-processed with v12 or later.

### Analyses

Any NanoAOD-based matching of the sub-leading photon finds nothing: HiggsDNA-style workflows, the HYgammaRunIII legend that lists bits 16 and 17 as the legs, FLAF configurations. MiniAOD-based flashgg reads the objects directly and is unaffected.

# How widespread (2): every filter bit of every object type, resolved against six menus

| object | bits | never settable (collection or type) | stale name in the 2025 menu | fine in 2025 |
|---|---|---|---|---|
| Electron | 20 | none | 2: WPLoose (gone 2023), triple-electron (gone 2025) | 18 |
| Photon | 21 | 4: bits 16, 17, 19 (unseeded) and 8 (hltHtEcal800) | 3: bit 12 (gone 2022), 18 and 19 (renamed 2023) | 15 |
| Muon | 14 | none | 2: mu-tau DeepTau overlap filters (gone 2025) | 12 |
| Tau | 31 | none | 1: DeepTau (gone 2025) | 30 |
| BoostedTau | 3 | v12 to v14: whole selection, see below | 1: 2022 ParticleNetTauTau name | 2 |
| Jet | 46 | none | 10: pixel-only HH4b filters (gone 2024), DeepCSV and DeepJet (gone 2025), 3-jet pt40 and pt45 | 34 (+2 L1 seeds) |
| FatJet | 7 | none | 3: 2022 and 2023 soft-drop and ParticleNet names | 4 |
| HT, MHT | 6, 2 | none | 2: Run 2 L1 seed names (never in Run 3 menus) | 3 (+1), 2 |
| MET | 0 | no bits with the skip flag on: MET objects never written | not applicable | none, all versions |

**Same mechanism elsewhere**

Only in the Photon list: the three unseeded diphoton bits and bit 8, whose filter saves HT objects. Electron and Muon have no collection mismatch. Two whole tables were empty by configuration: MET in every version since v9, and BoostedTau in v12 to v14, whose selection required a filter name ending in ParticleNetTauTau while the menus end in 0p30 or Tag0p03 (relaxed in 15_0_2).

**Stale names are the common case**

One bit list serves the 2022 to 2026 menus, so bits for filters that were renamed or retired stay documented but silently empty for later years, and bits added for new filters are empty for earlier years. That is a different, documentation-level problem; the photon one is a code defect that no menu can cure.

# Measured on real files: not one photon object with an unseeded bit, in any version since v12

| NanoAOD | file (one per dataset, read on lxplus) | events | fired Mass90 | photons: seeded bits | unseeded bits | e unseeded | id 2 | id 1515 |
|---|---|---|---|---|---|---|---|---|
| v9 | UL2018 data, EGamma 2018D | 300000 | 31771 | no photon TrigObj | none defined | n/a | 0 | 0 |
| v10 | 2022 prompt data, EGamma 2022D | 300000 | 46707 | bits not defined | none defined | n/a | 0 | 0 |
| v11 | 2022 prompt data, EGamma 2022F | 300000 | 41546 | bits not defined | none defined | n/a | 0 | 0 |
| v12 | Summer22 MC, GluGluHtoGG M125 | 12979 | 7725 | 15626 / 17978 | 0 / 0 / 0 | n/a | 0 | 0 |
| v12 | 2022 data, EGamma 2022D 22Sep2023 | 300000 | 46407 | 208527 / 255099 | 0 / 0 / 0 | n/a | 0 | 0 |
| v13 | Summer23 MC, GluGluHto2G M125 | 17760 | 10453 | 20865 / 24170 | 0 / 0 / 0 | 702 | 0 | 0 |
| v13 | 2022 data, EGamma 2022D 16Dec2023 | 300000 | 46794 | 207844 / 252774 | 0 / 0 / 0 | 29042 | 0 | 0 |
| v14 | 2024 prompt data, EGamma0 2024D | 300000 | 35892 | 203620 / 252039 | 0 / 0 / 0 | 19444 | 0 | 0 |
| v15 | Summer24 MC, GluGluH to 2G M125 | 4180 | 2406 | 4815 / 5588 | 0 / 0 / 0 | 205 | 0 | 22 |
| v15 | 2022 data, EGamma 2022D NanoAODv15 | 300000 | 48687 | 212270 / 258546 | 0 / 0 / 0 | 30451 | 0 | 430 |
| v15 | 2024 data, EGamma0 2024D MINIv6NANOv15 | 300000 | 30945 | 201628 / 248974 | 0 / 0 / 0 | 22980 | 0 | 254 |
| 133X | Winter24 MC, GluGluHToGG M125 powheg | 175654 | 101598 | 205090 / 239250 | 0 / 0 / 0 | 9860 | 0 | 0 |
| 16_0_X | 2026 data, EGamma0 2026C PromptReco | 300000 | 1511 | 124972 / 145161 | 0 / 0 / 0 | 1034 | 0 | 42 |

Seeded bits are the two leading-leg last filters (12 and 13 up to v14, 14 and 15 from v15); unseeded bits are the three sub-leading-leg filters (14, 15, 17 up to v14; 16, 17, 19 from v15). Every prediction holds: unseeded bits never set, seeded bits present in every diphoton-triggered event, the electron unseeded bit populated from v13, no MET objects (id 2) in any version, no BoostedTau objects (id 1515) before v15. Measurement of 4 November 2026, last three rows 5 November; results/REPORT.md on the notes branch.

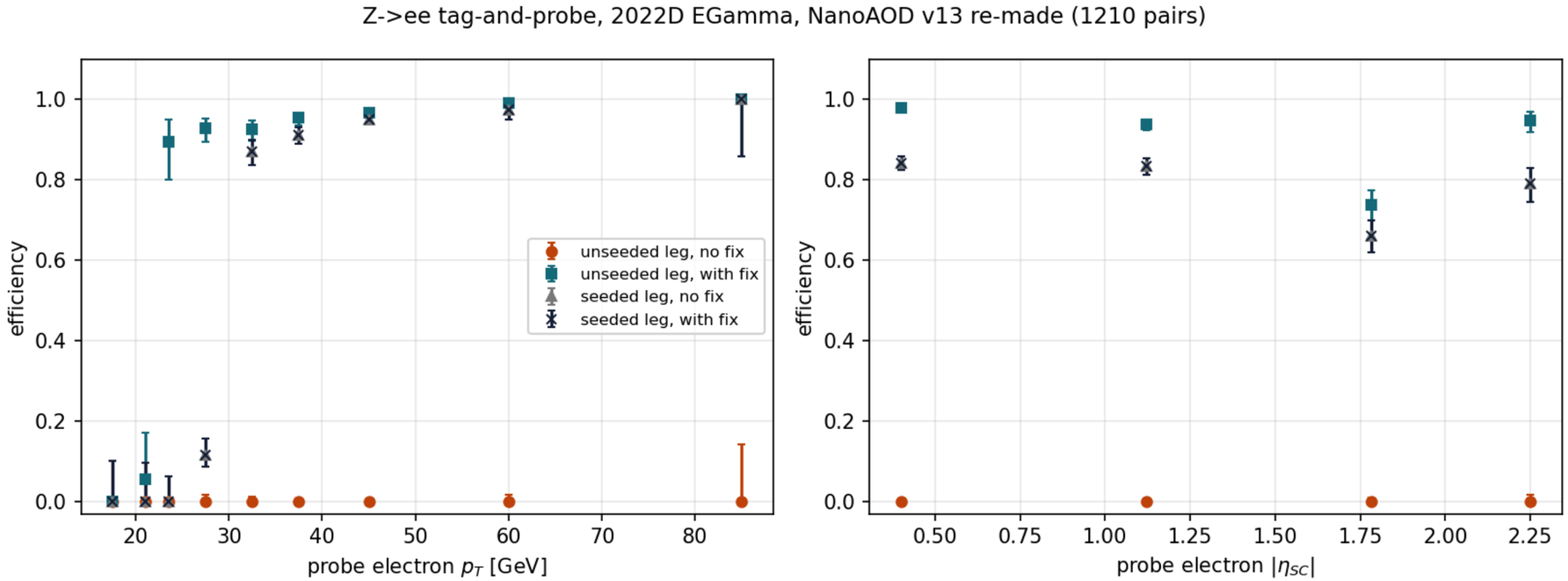
# Fix test on real input: every Mass90 event gets its sub-leading-leg bits, nothing else changes

| NANO file (v13, v15: 17760 events; 16_0_X: 20000) | fired Mass90 | photons: seeded bits | photons: unseeded bits | Mass90 events with an unseeded bit | nTrigObj mean | size (bytes) |
|---|---|---|---|---|---|---|
| central v13 file | 10453 | 20865 / 24170 | 0 / 0 / 0 | 0 | 5.287 | 29333713 |
| v13 re-made, no fix | 10453 | 20865 / 24170 | 0 / 0 / 0 | 0 | 5.287 | 29339920 |
| v13 re-made, with fix | 10453 | 20865 / 24170 | 20461 / 18365 / 0 | 10453 (all) | 5.388 | 29363499 |
| v15 re-made, no fix | 10453 | 20865 / 24170 | 0 / 0 / 0 | 0 | 38.134 | 39947152 |
| v15 re-made, with fix | 10453 | 20865 / 24170 | 20461 / 18365 / 0 | 10453 (all) | 38.225 | 39959608 |
| 16_0_X 2026C data, no fix | 308 | 8909 / 10760 | 0 / 0 / 0 | 0 | 2.100 | 26190407 |
| 16_0_X 2026C data, with fix | 308 | 8909 / 10760 | 671 / 1010 / 0 | 308 (all) | 2.118 | 26206694 |

- Re-made without the fix, the file is identical to the central v13 file: same TrigObj (id, bits, pt) in 17760 of 17760 events.
- The 21254 photon objects that gain an unseeded bit have pt from 22.04 GeV (median 58.30 GeV), none below the 22 GeV leg threshold.
- Every object type other than photons is unchanged in 17760 of 17760 events; photon objects 24835 to 26639 (v13), 25188 to 26807 (v15).
- The third unseeded bit (Diphoton30_18 leg) stays 0: that filter name does not exist in the 2023 menu; also 0 in 2026 data (not investigated).
- 2026 data, CMSSW_16_0_4_patch2, 20000 events of a Run2026C MINIAOD file: unseeded bits in all 308 Mass90 events, none before; pt from 22.01 GeV (median 37.06 GeV). Not compared event by event with a central file.
- Relval in CMSSW_16_1_3 with the fix (12452.0, 141.046): builds and runs without exceptions; outputs identical to the clean release except 25 DQM TimerService histograms per workflow. Its NANO TrigObj table is empty in both areas, so it does not exercise the photon selection.

Seeded / unseeded bits: 12, 13 / 14, 15, 17 in v13; 14, 15 / 16, 17, 19 in v15. Input: the MiniAOD parent of the central Run3Summer23 v13 GluGluHto2G M125 file; CMSSW_13_3_1_patch1 (v13) and CMSSW_15_0_15_patch4 (v15). 5 November 2026, results/REPORT.md "Fix test" and "Relval in CMSSW_16_1_3".

# Trigger efficiency with the fix: unseeded leg 0 to 0.93, seeded leg unchanged




| | no fix | with fix |
|---|---|---|
| unseeded leg (bit 14 or 15) | 0 / 1210 | 1125 / 1210 = 0.930 (0.922-0.937) |
| seeded leg (bit 12 or 13) | 979 / 1210 = 0.809 | 979 / 1210 = 0.809, identical in every bin |

Z to ee tag-and-probe on NANO re-made in CMSSW_13_3_1_patch1 from 2022D EGamma 22Sep2023 MINIAOD (62007 events); without the fix its TrigObj equals the central 16Dec2023 file in 19539 / 19539 shared events. Tag: tight electron, pT > 35 GeV, |eta_SC| < 2.1, matched to the HLT_Ele32_WPTight_Gsf object and to a seeded-leg photon object. Probe: loose electron, pT > 15 GeV, opposite charge, 70 < m_ee < 110 GeV (12 same-sign pairs). 68% Clopper-Pearson intervals. 2026C data (16_0_4_patch2, 20000 events): 42 pairs, unseeded leg 0 / 42 without and 39 / 42 with the fix.

# The fix: one line, mirroring what the Electron selection has done since 2023

```
PhysicsTools/NanoAOD/python/triggerObjects_cff.py, Photon = cms.PSet( ... )
- sel = cms.string("type(92) && pt > 15 && coll('hltEgammaCandidates')"),
+ sel = cms.string("type(92) && pt > 15 && (coll('hltEgammaCandidates') || coll('hltEgammaCandidatesUnseeded'))"),
```

### Why this form

It names exactly the two e/gamma collections and copies the Electron selection (PR 41827). A wildcard coll('hltEgammaCandidates*') would also match, but would silently admit any future collection with that prefix.

### What changes in the output

Unseeded objects enter the photon branch. With skipObjectsNotPassingQualityBits on, only those that set a photon bit are written, and the only photon-bit filters on the unseeded collection are the three diphoton ones. A seeded twin in the same direction is merged with OR-ed bits; otherwise a second photon entry appears, as for electrons.

### What does not change

Bit numbers, documentation strings, the other object types, the event content. Existing files cannot be repaired: the objects were never written. Only a new NANO pass with the fix restores the sub-leading leg.

**Status.** Commit 1699a39f7 on the fork branch claude/fervent-lamport-3oiuh6 and the same change as aebf3eb64 on top of upstream master ed35654b1 (branch nano-photon-trigobj-unseeded). Applies cleanly to today's master. Both branches are on the fork santanumahata/cmssw; nothing pushed to cms-sw/cmssw, no pull request opened yet.

# Validation, pull request, and the follow-ups this fix does not cover

## Validation

- Done 4-5 November: thirteen NanoAOD files, v9 to v15 and 2026 data (16_0_X), data and MC; unseeded photon bits 0 in every file.
- Done 5 November: NANO re-made with the fix (13_3_1_patch1, 15_0_15_patch4): unseeded bits in all 10453 Mass90 events, nothing else changed; 2026 data in 16_0_4_patch2: all 308 Mass90 events.
- Done 5 November: relval 12452.0 and 141.046 in CMSSW_16_1_3: no exceptions, outputs identical except DQM timing; relval NANO has no TrigObj.

## Pull request

- Branch nano-photon-trigobj-unseeded (commit aebf3eb64) is on the fork. Still to do: open it against cms-sw:master citing PR 40711 and PR 41827; xpog signature.
- Request a CMSSW_15_0_X backport if another NanoAOD production from that branch is planned.

## Follow-ups, separate from the one-line fix

- Add the 2023 names of the Diphoton30_18 filters as alternatives inside bits 18 and 19, so the numbering stays stable.
- Move or drop bit 8 (hltHtEcal800): its objects are HT objects and can never be photons.
- Review the stale bits per year (WPLoose, triple-electron, DeepTau overlaps, pixel-only jet filters, DeepCSV and DeepJet) and say in the documentation which menus each bit covers.
- Bits for the low-mass eta1p5, DiphotonMVA14p25, Diphoton15_1x EBEB and DoublePhoton paths, with the EGM POG and the Higgs PAG.
- Inform XPOG that MET trigger objects are never written and that the BoostedTau table was empty in v12 to v14.

Notes, results, patch, PR text: santanumahata/cmssw, branch nanoaod-trigobj-notes. Nothing pushed to cms-sw/cmssw.